

## Experiment: 9

### Massive MIMO Capacity vs SNR and Gain Using MATLAB

#### OBJECTIVE 1:

#### CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
snr = 0:5:30;           % SNR values in dB
```

```
snr_linear = 10.^(snr/10); % Convert dB to linear scale
```

```
capacity = log2(1 + snr_linear); % Shannon Channel Capacity
```

```
figure;
```

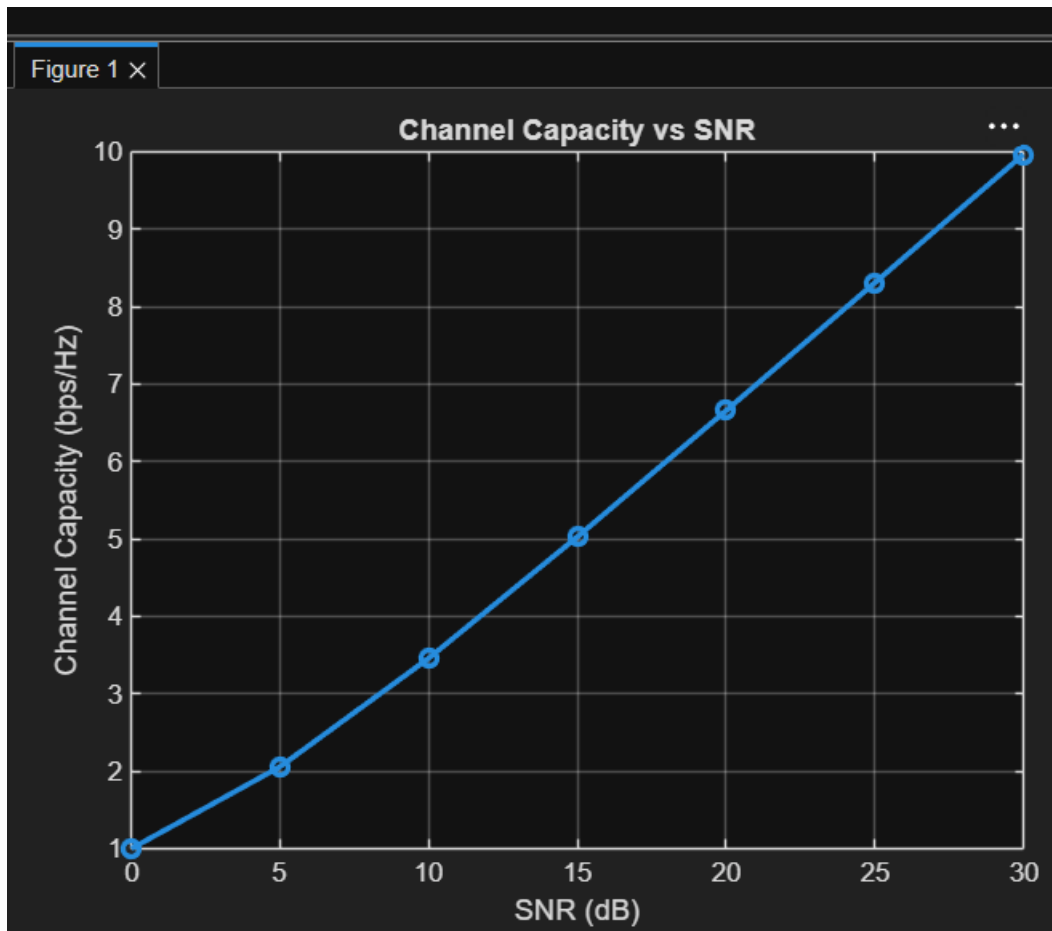
```
plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);
```

```
grid on;
```

```
xlabel('SNR (dB)');
```

```
ylabel('Channel Capacity (bps/Hz)');
```

```
title('Channel Capacity vs SNR');
```



OBJECTIVE 2:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
snr = 0:5:30;
```

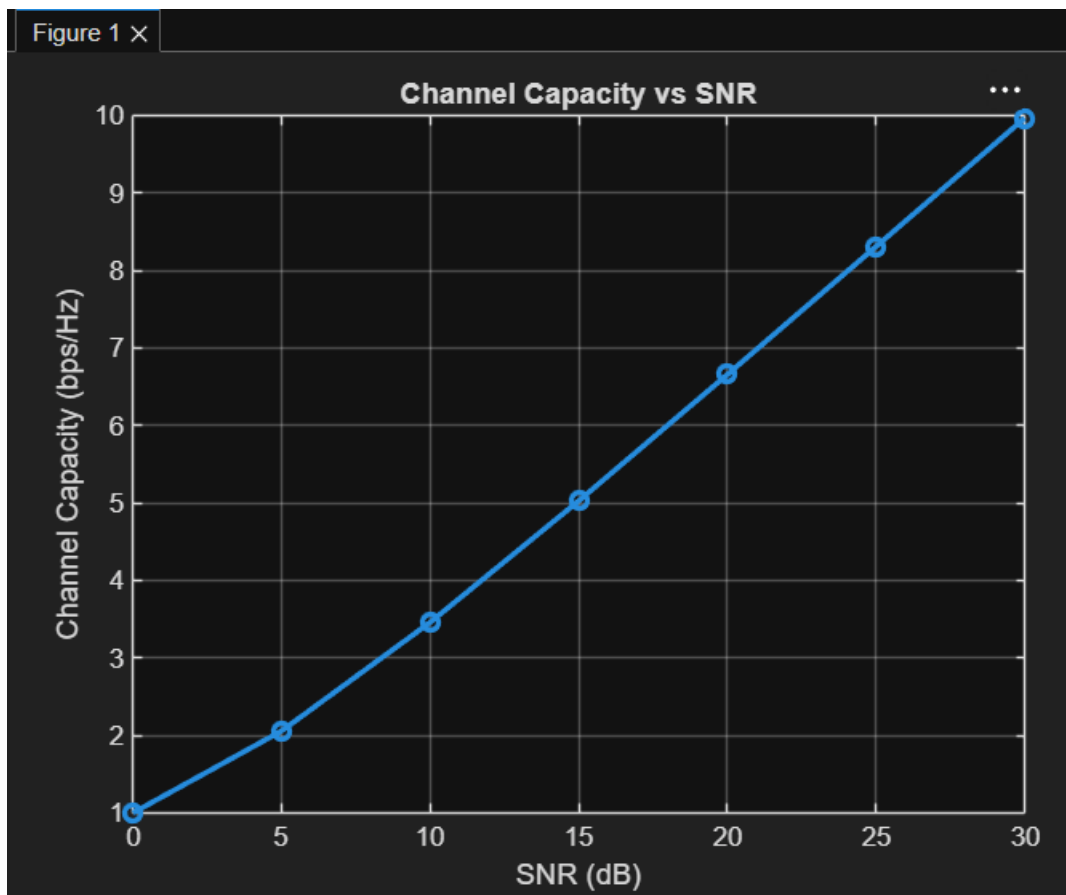
```
snr_linear = 10.^(snr/10);    % Convert SNR from dB to linear scale
```

```
capacity = log2(1 + snr_linear); % Shannon Capacity
```

```
figure;
```

```
plot(snr, capacity, '-o', 'LineWidth', 2, 'MarkerSize', 6);  
grid on;
```

```
xlabel('SNR (dB)');  
ylabel('Channel Capacity (bps/Hz)');  
title('Channel Capacity vs SNR');
```



OBJECTIVE 3:

CODE:

```
clc;  
clear;  
close all;
```

```
antennas = [4 8 16 32 64];    % Number of Antenna Elements  
gain = 10*log10(antennas);    % Antenna Gain (dB)
```

```
figure
```

```
bar(antennas, gain)
```

```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Antenna Gain (dB)')
```

```
title('Antenna Gain for Different Antenna Array Sizes')
```

